The times they are a-changin'" in Digital Humanities – a mini-conference on the temporal dimension of data

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Temporal Dynamics: Methodological Gaps and Opportunities in DH

Digital Humanities (DH) methods have significantly developed over recent decades. However, several blind spots still persist across the field, the insufficient attention to temporal dynamics in data being one of them.

Studies have demonstrated the need to account for the temporal dimension in data analysis, especially since some statistical methods output can be misleading without these adjustments (e.g., Koplenig 2015). This aligns with statistical approaches to time-series data, which advocate for modifying standard descriptive statistics to enable more accurate predictive analyses (Cryer and Chan 2009, 19).

Further, an important issue regarding temporal change is the methodological necessity of modeling: only if we understand the temporal aspects present in the training data can we effectively model the data to predict future trends (Bod 2018; Luo 2022). Not only is modeling the future intriguing, but the introduction of phylogenetic methods to corpus research has also provided new insights into the evolution of language, offering a framework for analyzing linguistic changes over time (da Silva und Tehrani 2016).

Temporal change, integral to a number of different disciplines with a DH focus, affects data, methodology, and analyses in a variety of ways: Digital art history and especially provenance research are challenged by changing or missing object information, such as varying titles (Kim 2015), or attributions (Hofbauer 2021), complicating access and tracing their (ownership) history over time and space. In geographical studies on mobility this movement in space and time is mediated by practices of sense-making (Creswell 2010). Discourse analysis examines time frames around key public events that set specific communicative strategies in motion (Islentyeva 2022), while corpus-based studies of registers, text varieties associated with the situation of use (Biber & Conrad 2019), are beginning to employ quantitative linguistic analyses to generate meaningful time periods (Gries & Hilpert 2008) to examine emergence and evolution of registers as cultural constructs. Digital Literary Studies raises questions about cultural evolution (Sobchuk 2023), genre history (Wagner-Egelhaaf 2014) and editorial histories of literary works (Bottigheimer 1987) on the level of texts, the evolution of authorial style in relation to the author (Piper 2018) and different types of time-series data such as eye movements and EEG data in empirical studies of reader-response (Dimigen u. a. 2011; Weitin u. a. 2024).

This raises critical questions: How should we address temporal change methodologically in the Digital Humanities? What role should visualizations play? Are time periods implemented to categorize data from the start (top-down), or are they the result of data-driven classification (bottom-up)? And how can we ensure that project data remains FAIR (Findable, Accessible, Interoperable, Reusable) long after the project concludes?

In the spirit of critical reflection on methodology in Digital Humanities (Meister 2012), we suggest a mini-conference format that explores various conceptualizations of time in DH projects, focusing on the domains of "images & objects", "text & language" and "place & space". This range aims to

highlight the challenges associated with the temporal dynamics in different types of data, including availability of temporal data in different disciplines and media, and identify common needs to be addressed by a shared methodology.

Organizers

The Department of Digital Humanities and Social Studies from FAU Erlangen-Nürnberg brings together a range of institutional and disciplinary perspectives, including art history, geography, linguistics and literary studies. Together, the mini-conference aims to address a shared methodological blind spot and to open a broader discussion on the topic of temporal change with prospectively a special issue in mind.

Target audience

The mini-conference welcomes researchers and practitioners from diverse disciplines and backgrounds who are engaged with the temporal dimensions of data. We encourage submissions from those involved in Digital Humanities, data analysis, historical studies, linguistic and literary research, art history, geo-spatial studies and related fields. By inviting a broad spectrum of voices, we aim to enrich our discussions and foster interdisciplinary dialogue that will deepen our understanding of time in the Digital Humanities.

Submission types and timeline

This mini-conference invites submissions for full-paper presentations (20 min. + 10 min. discussions), short paper/work-in-progress reports (15 min. + 15 min. discussions) from a variety of Digital Humanities domains.

Submissions can focus on but are not limited to the aforementioned domains ("images & objects", "text & language" and "place & space" etc.) and questions. Presentations can offer theoretical and methodological reflections, empirical studies, and/or practical applications on the topic of time in Digital Humanities.

The organizers expect submissions to include a presentation title, an abstract of max. 250 words, (bibliography excluded), and the chosen format of the presentation (full paper / short paper). The preliminary submission deadline will be on April 1st, 2025, while acceptance notifications will be sent out by April 15, 2025. We plan to issue a Call for Papers after the acceptance notices for the DH conference are published in late February 2025.

Preliminary schedule

The schedule suggested below is provisional and will be adjusted according to the profile of submissions as well as the proposed time slots for coffee and lunch. The lightning talks in the beginning and the conference wrap-up in the end as well as each panel discussion chair are covered by the organizers to ensure that the exchange is steered into a fruitful direction.

Time	Session	Content
10:00-11:00	Perspectives on and challenges of change in DH	Lightning talks by the organizers: • Images and objects • Text and language • Place and Space
11:00-11:15	Coffee break	
11:15-12:45	Text and Language	Accepted talks (Chairs: Anastasia Glawion / Marianna Gracheva)
12:45-13:45	Lunch break	
13:45-15:15	Images and objects	Accepted talks (Chair: Sabine Lang)
15:15-15:30	Coffee break	
15:30-17:00	Space and Place	Accepted talks (Chair: Dominik Kremer)
17:00-17:45	Discussion and concluding remarks	Wrap up/social, open-end networking

Bibliography

Biber, Douglas / Conrad, Susan (2019): Register, genre, and style. Cambridge: Cambridge University Press.

Bod, Rens (2018). "Modelling in the Humanities: Linking Patterns to Principles", in: Historical Social Research / Historische Sozialforschung 31: 78–95.

Bottigheimer, Ruth B. (1987): "Silenced Women in the Grimm's Tales: The ‚Fit' between Fairy Tales and Society in Their Historical Context", in: Bottigheimer, Ruth (ed.): Fairy Tales and Society: Illusion, Allusion, and Paradigm. Philadelphia: University of Pennsylvania Press 115–133. DOI: 10.9783/9780812201505.

Cresswell, Tim (2010): “Towards a Politics of Mobility“, in: Environment and Planning D: Society and Space, 28, 1: 17-31. DOI: 10.1068/d11407

Cryer, Jonathan D. / Kung-Sik Chan (2008): Time Series Analysis: With Applications in R. New York, NY: Springer.

Dimigen, Olaf / Sommer, Werner / Hohlfeld, Annette / Jacobs, Arthur M. / Kliegl, Reinhold (2011) “Coregistration of Eye Movements and EEG in Natural Reading: Analyses and Review“, in: Journal of Experimental Psychology 140, 4: 552–72. DOI: 10.1037/a0023885.

Gries, Stephan Th. / Hilpert, Martin (2008): “The identification of stages in diachronic data: variability-based neighbour clustering”, in: Corpora 3,1: 59-81. DOI: 10.3366/E1749503208000075.

Hofbauer, Michael (2021): Cranach: Parerga und Paralipomena: Neues zu Lucas Cranach und seinen Söhnen. Heidelberg: arthistoricum.net. DOI: 10.11588/arthistoricum.722.

Islentyeva, Anna (2023): „British Media Representations of EU Migrants before and After the EU Referendum”, in: Critical Approaches to Discourse Analysis across Disciplines 14, 3: 1-20.

Kim, Sukmo (2015): Bildtitel: eine Kunstgeschichte des Bildtitels. Hamburg: Kovač.

Koplenig, Alexander (2017): “Why the Quantitative Analysis of Diachronic Corpora That Does Not Consider the Temporal Aspect of Time-Series Can Lead to Wrong Conclusions“, in: Digital Scholarship in the Humanities 32, 1: 159-168. DOI: 10.1093/llc/fqv030.

Luo, Sha (2022): “Forecasting fund-related textual emotion trends on Weibo: A time series study“, in: Frontiers in Communication 7: 970749. DOI: 10.3389/fcomm.2022.970749.

Piper, Andrew (2018): Enumerations: data and literary study. Chicago, IL; London: The University of Chicago Press.

Silva, Sara Graça da / Jamshid J. Tehrani (2016): “Comparative Phylogenetic Analyses Uncover the Ancient Roots of Indo-European Folktales“, in: Royal Society Open Science 3,1: 150645. DOI: 10.1098/rsos.150645.

Sobchuk, Oleg (2023): “Evolution of Modern Literature and Film“, in: Tehrani, Jamshid / Kendal, Jeremy / Kendal, Rachel (eds.): The Oxford Handbook of Cultural Evolution. Oxford: Oxford University Press. DOI: 10.1093/oxfordhb/9780198869252.013.45.

Wagner-Egelhaaf, Martina (2014): “Literaturgeschichte als operative Fiktion“, in Matthias Buschmeier (ed.): Literaturgeschichte: Theorien - Modelle - Praktiken. Studien und Texte zur Sozialgeschichte der Literatur. Berlin: De Gruyter 86–101.

Weitin, Thomas / Fabian, Thomas / Glawion, Anastasia / Brottrager, Judith / Pilz, Zsofia (2024): “Is badfiction processed differently by the human brain? An electrophysical study on reading experience“, in: Frontiers in Human Neuroscience 17: 1333965. DOI: 10.3389/fnhum.2023.1333965.